CPU EDUCATION

COMP90051 期末班

5 讲解

TUTOR：JJ 学长

时长：2 小时

本周内容预览：

1.BN 和 MRF 的表达式和条件独立性

2.BN 和 MRF 的 Junction Tree 形成

3.BN 和 MRF 的 learning

CPU EDUCATION

关于CPU EDUCATION

CPU Education成立于2018年，总部在澳大利亚墨尔本，目前国内办公室在广州。经过近多年的积累与发展，公司凭借优质的教学质量和专业的研发团队力量不断壮大，成为全球最受学生欢迎的教育机构之一，在众多学生及老师中享有良好的口碑。CPU Education争做行业标杆，为不同阶段的学生提供专业，专属，专注的教学服务。

致力于为学生实现人人H1的目标，以“为学生提供极具价值的教学服务，培训各行业精英”为宗旨。秉承“教育高于一切”的经营理念，本着服务定制化、师资专业化、教学规范化的要求，打造墨尔本教育行业的“黄埔军校”。

课后，如果您有任何建议和意见，我们都非常欢迎您联系小助手分享您的想法，给予我们改进和提高的机会!
感谢您参与CPU Education的辅导课程!

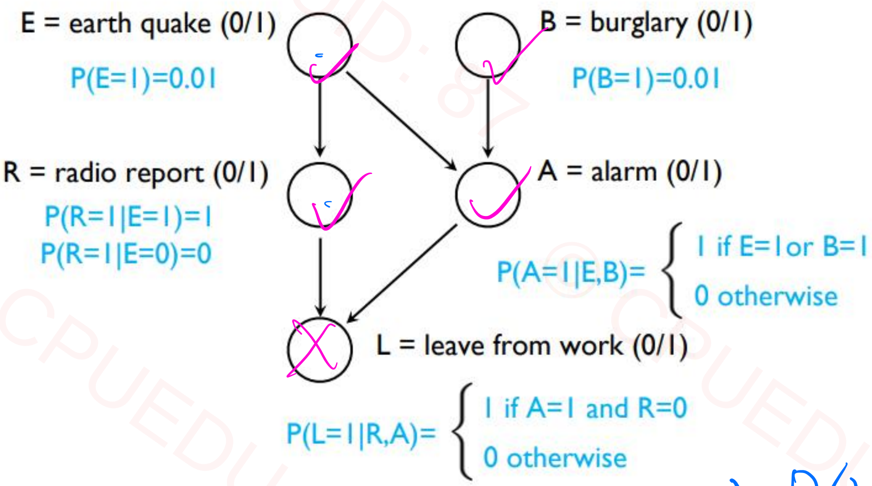
课程反馈

PART 1 BN 和 MRF

DAG

A BN defines the joint distribution by:

$$P(X_1,\ldots,X_n)=\prod_{i=1}^{n}P(X_i\mid \mathrm{Pa}(X_i)).$$



$P(E)\cdot P(B)\cdot P(R|E)\cdot P(A|B,E)\ P(L|R,A)$

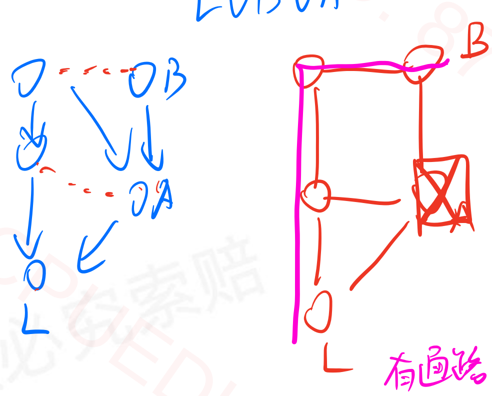
1.4 Conditional Independence via D-Separation

To test if $\boxed{X \perp Y \mid S}$ in a BN:

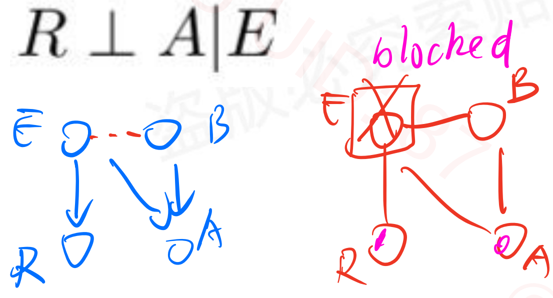
1. Build the **ancestral subgraph** of $X \cup Y \cup S$.
2. **Moralize** it: connect co-parents and drop directions.
3. Apply **ordinary graph separation**: if all undirected paths between X and Y are blocked by S, then $X \perp Y \mid S$. x,Y路 blocked

$L \perp B|A$ X

$L \cup B \cup A$

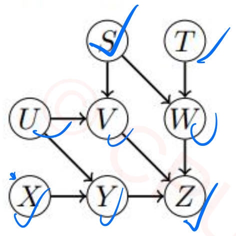


$R \perp A|E$



$R \perp A|E$ ✓

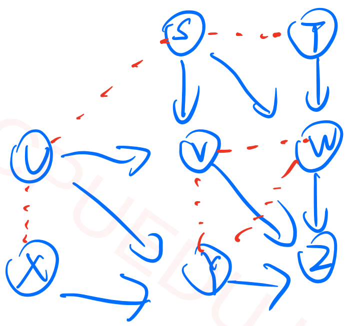
Question 1. Consider the following Bayesian network:



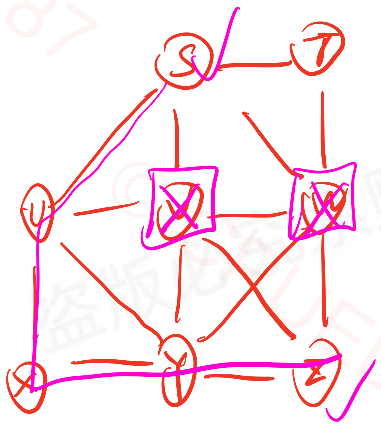
Does the conditional independence statement $S \perp Z \mid V, W$ hold in this Bayesian network?

① 找 Ancestral subgraph

$S \cup Z \cup V \cup W$



② Moralize

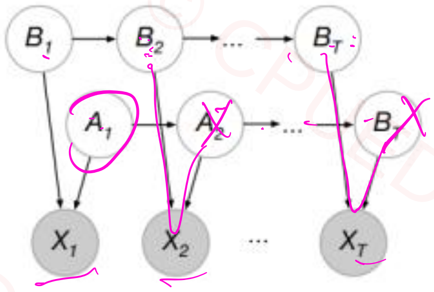


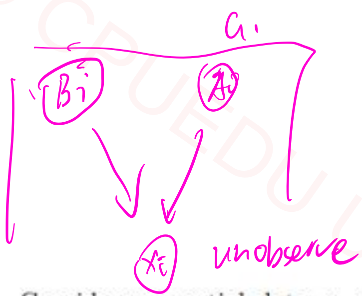
③ V, W 连接

S, Z 存在通路

$S \perp Z \mid V, W$ 不成立

An alternative to the single chain *hidden Markov model* is the multiple chain variant, illustrated below:





unobserve

Consider sequential data $x_1, x_2, \ldots, x_T$ of T symbols, and two hidden chains with hidden states $a_t, b_t, t \in \{1, \ldots, T\}$, respectively. All random variables are binary valued, i.e., $x_t, a_t, b_t \in \{0, 1\}, t \in \{1, \ldots, T\}$. Chains are *homogenous*, i.e., the probability of transitions and generations are independent of time step, t.

a) What random variables are *marginally independent* of a_1? As we are considering marginal independence, no values of x_t are observed. Explain your answer. [6 marks]

$B_1, B_2 \cdots B_T$ 都和 a_1 独立

因为在 x_i unobserve 时，

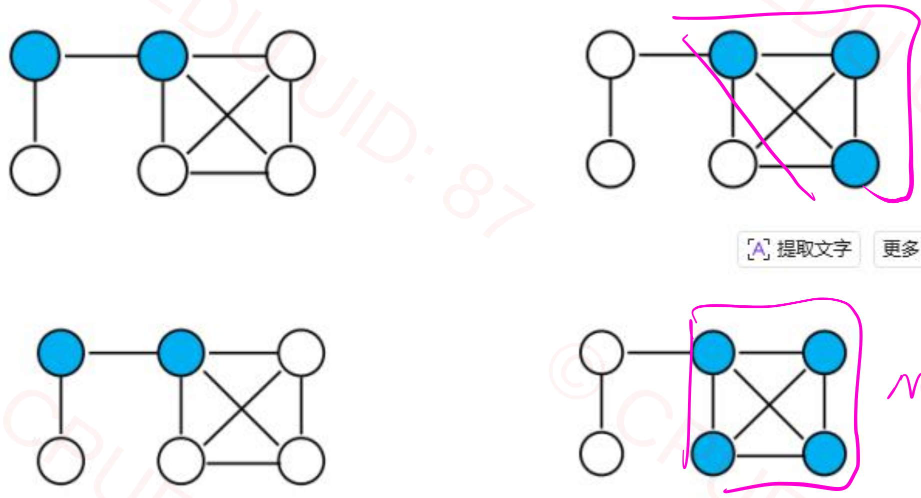
Ai和Bi是独立的

MRF

2.2 Undirected Graphs and Cliques

无向图

- An **undirected graph** links nodes with undirected edges.
- A **clique** is a fully connected set of nodes; a **maximal clique** is not a subset of a larger clique.



包含

maximum clique

☆☆☆ Let $\psi_C(X_C) > 0$ be a potential for each (maximal) clique C. The joint is:

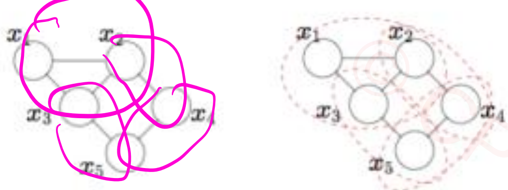
$$P(X) = \frac{1}{Z}\prod_C \psi_C(X_C), \qquad Z = \sum_x \prod_C \psi_C(x_C).$$

Theorem: (Hammersley-Clifford) Any distribution that is consistent with an undirected graph has to factor according to the (maximal) cliques in the graph

$$P(x_1, \ldots, x_n) = \frac{1}{Z}\prod_{c\in\mathcal{C}} \psi_c(x_c)$$

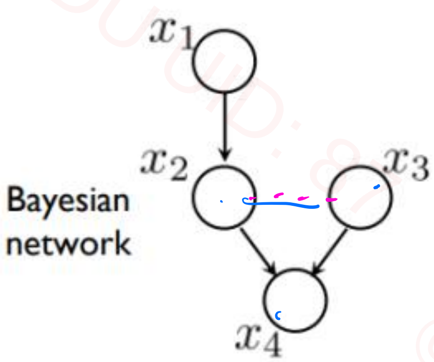
表达为所有最大势乘积再除以Z

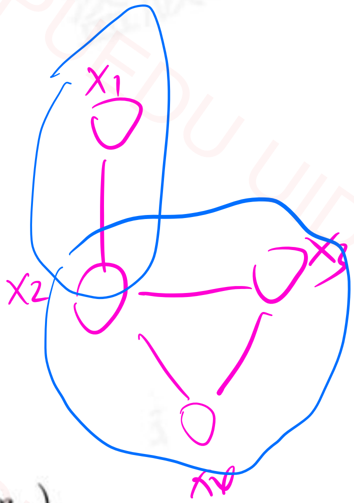
where $x_c = \{x_j, j \in c\}$ denotes the variables in clique c. $\psi_c(x_c)$ is any positive valued function of the variables contained in clique c (clique potential).



2.5 Mapping BNs to MRFs

1. Take the BN over the variables of interest.
2. Moralize: connect all co-parents; remove edge directions.
3. The resulting undirected graph defines the MRF structure.

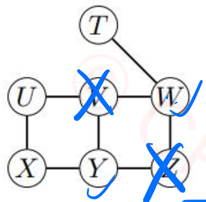




$$P(x_1)P(x_2|x_1)P(x_3)P(x_4|x_2,x_3)$$

$\frac{1}{Z} \cdot \psi_{12}(x_1, x_2) \cdot \psi_{234}(x_2, x_3, x_4)$

Question 2. Consider the following Markov random field:



Does the conditional independence statement $W \perp Y \mid \boxed{V, Z}$ hold in this Markov random field?

Blocked

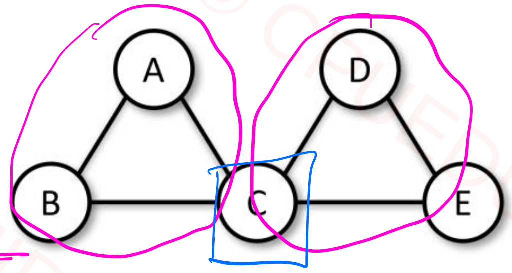
$W \perp Y \mid V, Z$

Question 2: Probabilistic Graphical Models [14 marks]

MRF

Consider the following *undirected probabilistic graphical model (U-PGM)* on five Boolean-valued variables, where the *clique potentials* $f(A, B, C)$ and $G(C, D, E)$ are shown below.

A	B	C	f(A,B,C)
T	T	T	9
T	T	F	3
T	F	T	4
T	F	F	3
F	T	T	5
F	T	F	3
F	F	T	2
F	F	F	3



C	D	E	g(C,D,E)
T	T	T	0
T	T	F	0
T	F	T	0
T	F	F	0
F	T	T	2
F	T	F	4
F	F	T	4
F	F	F	1

(a) Using the given *clique potential tables*, calculate the *normalising constant* (aka *partition function*) for the joint distribution on A, B, C, D, E. [8 marks] Z

(b) Calculate $\Pr(A = F, B = F, C = F)$. You may leave your answer as a fraction. (If you were unable to answer the previous part, leave the constant as Z in your workings here.) [6 marks]

c).

$$Z = \sum_{A,B,C,D,E} f(A,B,C)\, g(C,D,E)$$

$$= 3\times2 + 3\times4 + 3\times4 + 3\times1$$
$$+ 3\times2 + 3\times4 + 3\times4 + 3\times1$$
$$+ 3\times2 + 3\times4 + 3\times4 + 3\times1$$
$$+ 3\times2 + 3\times4 + 3\times4 + 3\times1$$
$$= 132$$

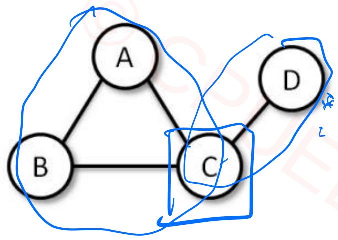
b).

$$\Pr(A=F, B=F, C=F) = \frac{1}{Z} \cdot \sum_{D,E} f(A=F, B=F, C=F) \cdot g(C=F, D, E)$$
$$= \frac{1}{132} \cdot 3 \cdot (2+4+4+1)$$
$$= \frac{33}{132}$$

Question 8: Probabilistic Graphical Models [4 marks]

Consider the following *undirected probabilistic graphical model (PGM)* on four Boolean-valued variables, where the *clique potentials* $f(A, B, C)$ and $g(C, D)$ are unnormalised.

A	B	C	f(A,B,C)
T	T	T	3
T	T	F	1
T	F	T	1
T	F	F	6
F	T	T	2
F	T	F	4
F	F	T	4
F	F	F	0



C	D	g(C,D)
T	T	3
T	F	0
F	T	0
F	F	3

1. Using these tables, calculate the *normalising constant* $Z = \sum_{A\in\{T,F\}} \sum_{B\in\{T,F\}} \sum_{C\in\{T,F\}} \sum_{D\in\{T,F\}} f(A, B, C)g(C, D)$ [2 marks]
2. Calculate $\Pr(A = F, B = T, C = F)$. You may leave your answer as a fraction. (If you were unable to answer the previous part, leave the constant as Z in your workings here.) [2 marks]

1). $Z = 3\times(3+1+2+4) + 3\times(1+6+4+0)$

$= 63$

2). $\Pr(A=F, B=T, C=F) = \frac{1}{Z}\sum_{D} f(A=F, B=T, C=F)\, g(C=F, D)$

$= \frac{1}{63}\cdot 4\cdot(0+3)$

$= \frac{12}{63}$

PART 2 Inference

$$P(x_i) = \sum_{x_{-i}} P(x_1, x_2, \ldots, x_n)$$

Inference 求解 Node 边际分布
或 Given 节点条件分布

$$P(x_i | e) = \frac{P(x_i, e)}{P(e)}$$

Y \ X	0	1
0	0.1	
1	0.2	

$P(X=0) = 0.1 + 0.2 = 0.3$

Variable elimination is a procedure for computing marginals by summing out variables one at a time. Each elimination step combines local factors and creates a new factor summarizing the influence of the eliminated variable.

For a joint distribution factorized as

$$P(x_1, \ldots, x_n) = \frac{1}{Z} \prod_{(i,j) \in E} \psi_{ij}(x_i, x_j),$$

we can eliminate a variable x_k by computing:

$$\phi'(x_{N(k)}) = \sum_{x_k} \prod_{j \in N(k)} \psi_{kj}(x_k, x_j),$$

where $N(k)$ denotes the neighbors of node k. The resulting factor $\phi'(x_{N(k)})$ is then passed to its neighbors.

Variable elimination can be interpreted as a process of passing messages between nodes in the graph: each node sends a message to its neighbor summarizing how its local factors and its sub-tree influence the neighbor.

The message from node i to node j is: $\boxed{X_i} \to X_j$

$$\boxed{m_{i \to j}(x_j)} = \sum_{x_i} \psi_{ij}(x_i, x_j) \prod_{k \in N(i) \setminus j} m_{k \to i}(x_i).$$

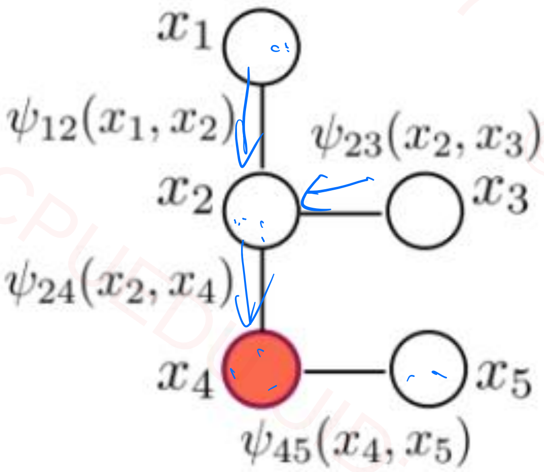
其它节点到 x_i 的 message

This message represents how much node i supports each possible state of node j. Once all incoming messages to a node are received, its marginal can be computed as:

$$P(x_i) \propto \prod_{k \in N(i)} m_{k \to i}(x_i),$$

with normalization ensuring

$$\sum_{x_i} P(x_i) = 1.$$



$m_{1\to2}(x_2) = \sum_{x_1} \psi_{12}(x_1, x_2)$

$m_{2\to4}(x_4) = \sum_{x_2} \psi_{24}(x_2, x_4)\, m_{1\to2}(x_2)\, m_{3\to2}(x_2)$

$P(x_4) \propto m_{2\to4}(x_4) \cdot m_{5\to4}(x_4)$

Belief Propagation : Tree (不能处理环)

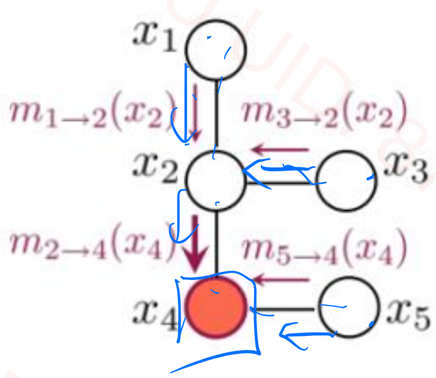
3.2 Two Phases of BP

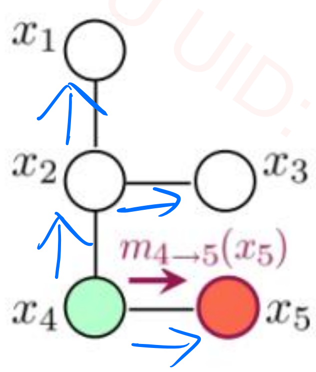
1. **Collect (Upward) Phase:** Messages are propagated from the leaves toward a chosen **root node**. Each node summarizes information from its sub-tree and sends it to its parent.

2. **Distribute (Downward) Phase:** After the root collects all messages, it sends information back down the tree, allowing every node to compute its marginal.

At the end of the collect phase, the root node has aggregated information from the entire graph. If we only need the marginal of a specific node, that node can simply be treated as the root.

因为这个算法有来回的双向信息计算





3.3 BP Rules

- **Message Update:** Outgoing message = product of incoming messages and local pairwise potential, summed over the sender's variable.

$$m_{i\to j}(x_j) = \sum_{x_i} \psi_{ij}(x_i, x_j) \prod_{k\in N(i)\setminus j} m_{k\to i}(x_i)$$

- **Node Marginal:**

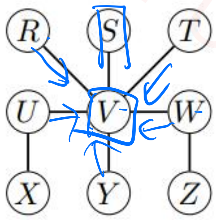
$$P(x_i) \propto \prod_{k\in N(i)} m_{k\to i}(x_i)$$

$P(x_2, x_4) \propto \psi_{24}(x_2, x_4)\, m_{3\to 2}(x_2)$
$m_{1\to 2}(x_2)$
$m_{5\to 4}(x_4)$

- **Pairwise Marginal:**

$$P(x_i, x_j) \propto \psi_{ij}(x_i, x_j) \prod_{k\in N(i)\setminus j} m_{k\to i}(x_i) \prod_{l\in N(j)\setminus i} m_{l\to j}(x_j)$$

Question 3. Consider the following Markov random field:



After running the Belief Propagation algorithm, how should we correctly compute the marginal distribution $P(V)$?

$P(V) \propto m_{R\to V}(V)\, m_{S\to V}(V)\, m_{T\to V}(V)\, m_{W\to V}(V)\, m_{U\to V}(V)$
$m_{Y\to V}(V)$

Assume we want to send a message from V to S. How should we correctly compute the message $m_{V\to S}(S)$?

$\sum_V \psi_{SV}(S, V) \cdot m_{R\to V}(V) \cdot m_{T\to V}(V) \cdot m_{W\to V}(V)$
$\cdot m_{U\to V}(V) \cdot m_{Y\to V}(V)$

4.1 Steps to Form a Junction Tree 针对于 General Graph

步骤

The process of forming a junction tree from an original graphical model involves the following key steps:

1. **Moralization:** Convert the directed graphical model into an undirected one by connecting all parents of each node (to "marry" them) and then dropping the direction of all edges. 有向图→无向图

2. **Triangulation (Variable Elimination Ordering):** Add additional edges to ensure that every cycle of length greater than three has a chord (an edge connecting non-adjacent nodes in the cycle). The elimination order determines how cliques are formed. Finding the optimal order is NP-hard, but good heuristics exist. 决定一个消去顺序，每消去一个，连接消去节点父母，并写出包含消去点的Maximum clique

3. **Clique Formation:** Identify all maximal cliques in the triangulated graph. Each clique represents a cluster of variables that will be a node in the junction tree.

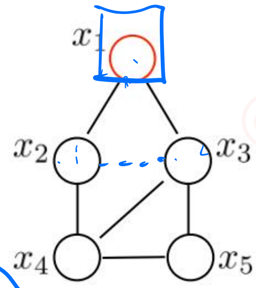
4. **Junction Tree Construction:** Connect the cliques to form a tree structure such that for any variable that appears in multiple cliques, all those cliques are connected — this is known as the *running intersection property*. 找每个Maximum Clique当一个超节点，形成一个最大权重树

5. **Message Passing (Inference):** Perform sum-product (or max-product) message passing along the junction tree to compute marginals. The computational complexity depends on the size of the largest clique. 用BP做Inference

Complexity Insight:

- The complexity of inference in the junction tree depends exponentially on the size of the largest clique.
- The elimination order strongly affects the clique sizes and thus the efficiency of inference.
- A common heuristic is to eliminate the variable that produces the smallest resulting clique at each step.

② Elimination order: $x_1\ x_2\ x_3\ x_4\ x_5$



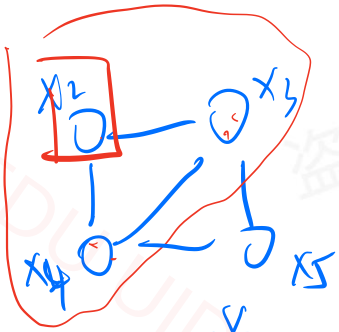
① Moralize

已经 Moralize

消去X1

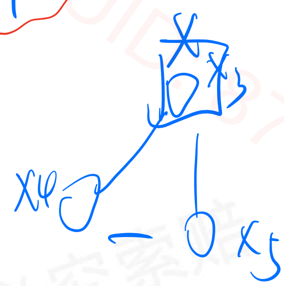
$C_1=\{X_1,X_2,X_3\}$

消去X2



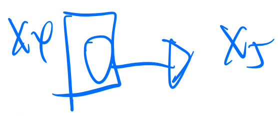
$C_2=\{X_2,X_3,X_4\}$

消去X3



$C_3=\{X_3,X_4,X_5\}$

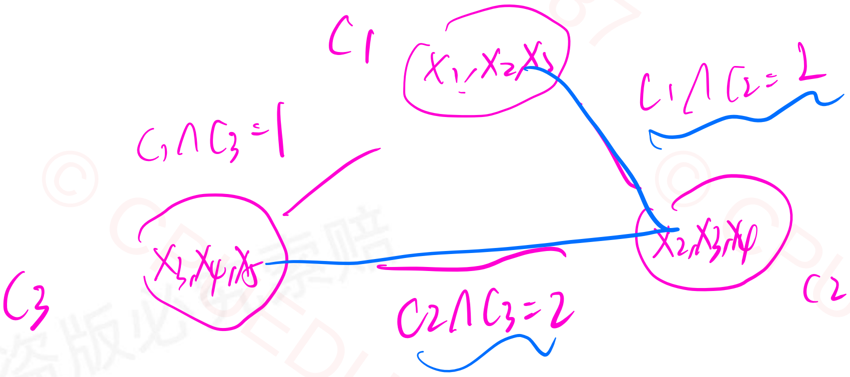
消去X4



$C_4=\{X_4,X_5\}$

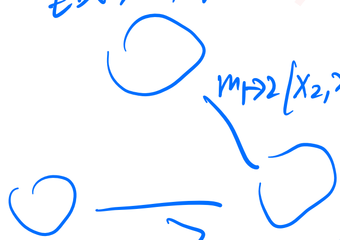
消去X5

$\{X_5\}$



生成无环最大权重树

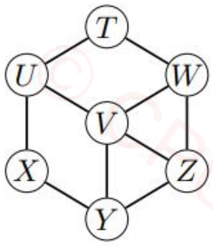
BP



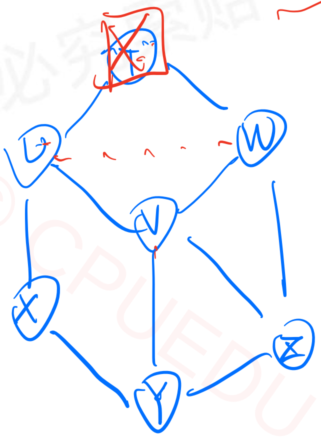
$m_{1\to2}(X_2,X_3)=\sum_{X_1}\psi_1(X_1,X_2,X_3)$

$m_{3\to2}(X_3,X_4)=\sum_{X_5}\psi_3(X_3,X_4,X_5)$

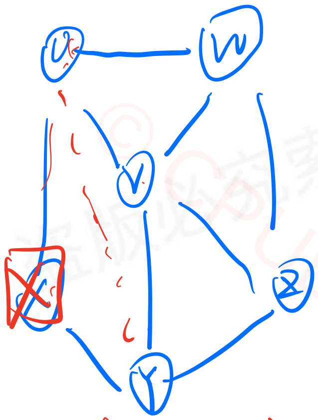
Question 4. Consider the following Markov Random field:



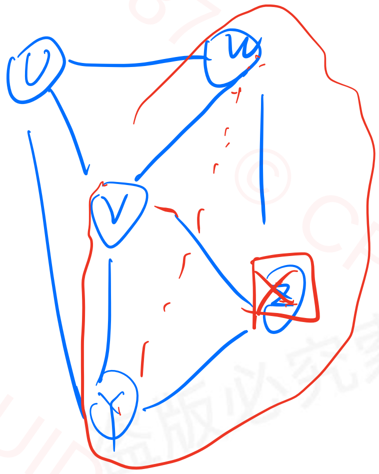
Assume the following node elimination order: T, X, Z, U, W, V, Y. (Eliminate the first node in the list, then eliminate the second node in the list, then eliminate the third node in the list, etc.) What are the cliques in the junction tree (by using the above elimination order)?



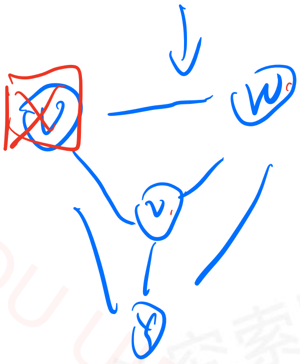
$C_1(T,U,W)$



$C_2(U,X,Y)$

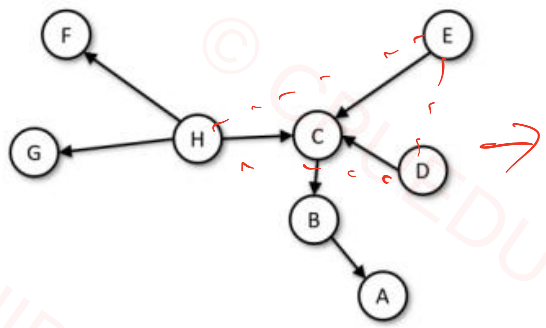


$C_3=\{Z,Y,V,W)$

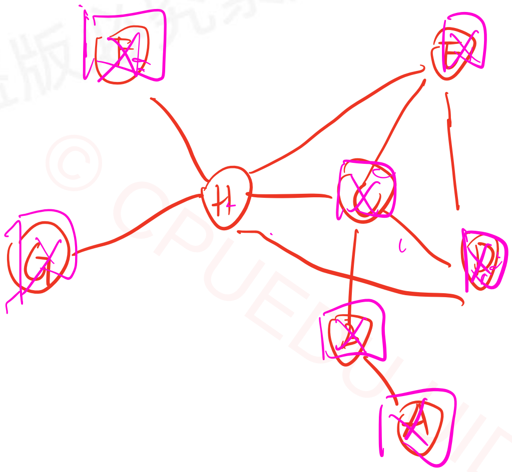


$C_4\{W,V,Y,U)$

Consider the following *directed probabilistic graphical model* (PGM) with Boolean-valued random variables



What is the maximum clique size for the junction tree for elimination order ABCDEFG? 形成Junction Tree



$C_1 = \{A, B\}$ ✓

$C_2 = \{B, C\}$ ✓

$C_3 = \{C, D, E, H\}$ ✓

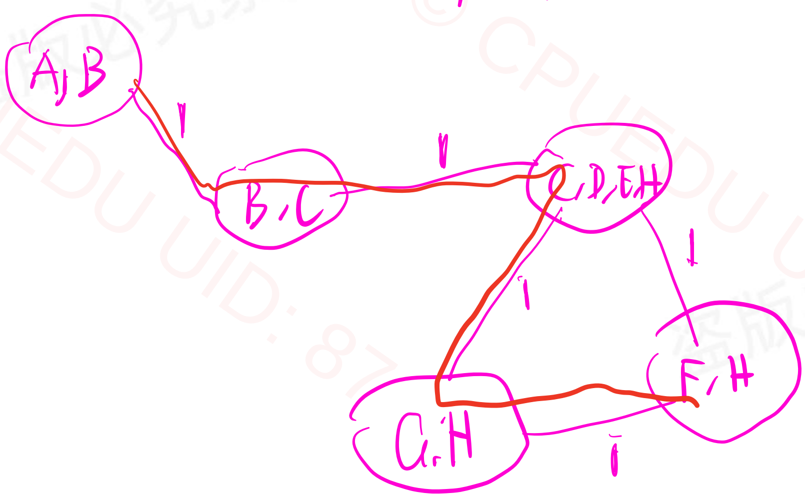
$C_4 = \{D, E, H\}$ X

$C_5 = \{E, H\}$ X

$C_6 = \{F, H\}$ ✓

$C_7 = \{G, H\}$ ✓

Maximum clique Size: 4



PART 3 PGM learning

BN 学习的是结构和参数

p(x|parent) 利用MLE (频率)

$$D = \begin{matrix} 2 & 2 & 1 & 1 & 1 & 2 & 1 & 1 & 2 & 1 & 2 \\ 1 & 2 & 2 & 2 & 2 & 2 & 3 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 2 & 1 & 1 & 1 & 1 & 1 & 1 \\ 2 & 2 & 3 & 1 & 1 & 2 & 3 & 3 & 2 & 3 & 2 \\ 1 & 2 & 2 & 2 & 2 & 2 & 3 & 1 & 1 & 2 & 2 \\ 1 & 1 & 3 & 1 & 1 & 1 & 3 & 1 & 1 & 1 & 3 \\ & & & & & \cdots & & & & & \end{matrix}$$

complete data

parameter estimation

+ complexity penalty

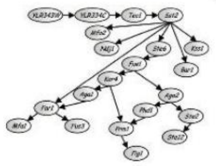
$$\text{score}(G; D) = \sum_{i=1}^{n} \text{score}(i|pa_i, D)$$

likelihood + 限制模型 complexity

decomposable scoring function

每个图的score分解为每个条件分布的score加和

structure search



$$\arg\max_G \ \text{score}(G; D)$$

highest scoring acyclic graph

Refined Bayesian score

Q1 MLE likelihood 为什么不能用于选择父节点结构？
因为它只能判别对数据的拟合好坏，
因为加更多节点，似然性总是增加

Q2 BIC score 和 likelihood 差了什么？
多了一个考虑自由参数数量和样本数的惩罚项

Q3 Refined Bayesian score 和原本 score 区别
由似然性+penalty
单一θ (最大似然值)

不只看单一θ
而是对所有θ加权求和
(期望)

MRF 计算似然

Parameter estimation for trees

- Suppose we evaluate counts $\hat{n}_{ij}(x_i, x_j)$ from T complete observations for any pair of variables x_i and x_j as well as counts $\hat{n}_i(x_i)$ for any individual variable x_i. As a result, we have the corresponding empirical probability estimates

$$\hat{P}_{ij}(x_i, x_j) = \frac{\hat{n}_{ij}(x_i, x_j)}{T}, \quad \hat{P}_i(x_i) = \frac{\hat{n}_i(x_i)}{T}$$

(moment matching)

- A *tree structured distribution* for edges $(i, j) \in E$ included in the tree can be written as

$$P(x_1, \ldots, x_n) = \left[\prod_{i=1}^{n} P_i(x_i)\right] \prod_{(i,j)\in E} \frac{P_{ij}(x_i, x_j)}{\boxed{P_i(x_i)P_j(x_j)}}$$

- These are valid maximum likelihood parameter estimates for the tree

How does $\mathbb{E}_{x_1,\ldots,x_n\sim P}[\log P(x_1, \ldots, x_n)]$ look like?

学习的是 MRF 结构

Comparing different trees

不管结构怎么变
只要Data不变
自信息是一样的

- The resulting value of the log-likelihood can be written as

自信息 互信息

$$l(D; \hat{\theta}) = T\left[-\sum_{i=1}^{n} \hat{H}(X_i) + \sum_{(i,j)\in E} \hat{I}(X_i; X_j)\right]$$

where $\hat{H}(X_i)$ is the empirical estimate of the entropy of x_i, and $\hat{I}(X_i; X_j)$ is the empirical estimate of the mutual information between the corresponding variables:

$$\hat{H}(X_i) = -\sum_{x_i=1}^{r_i} \hat{P}_i(x_i) \log \hat{P}_i(x_i)$$

$x_i \in \{1, \ldots, r_i\}$

$$\hat{I}(X_i; X_j) = \sum_{x_i=1}^{r_i} \sum_{x_j=1}^{r_j} \hat{P}_{ij}(x_i, x_j) \log \frac{\hat{P}_{ij}(x_i, x_j)}{\hat{P}_i(x_i)\hat{P}_j(x_j)}$$

- "Best scoring tree" therefore should include edges with high mutual information. (The sum of H's is the same for all trees)

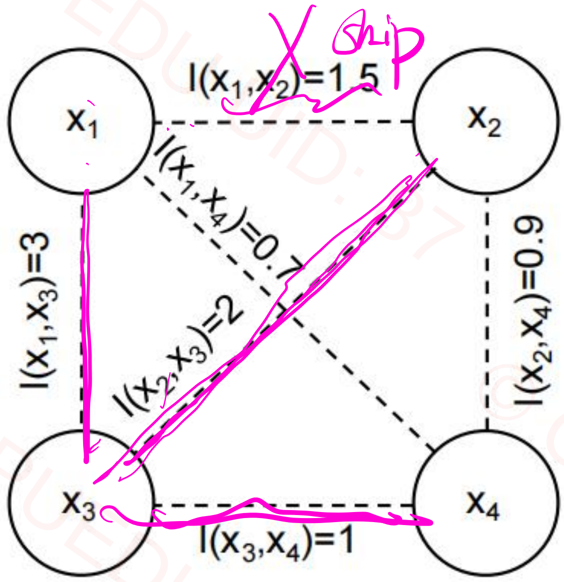
当计算好了边的互信息

4.3 Maximum Weighted Spanning Tree (MWST)

The optimal tree structure can be efficiently found using the MWST algorithm:

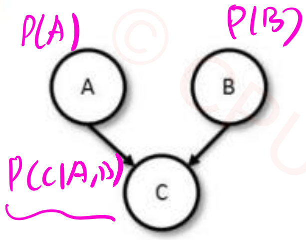
1. Treat each variable as a node.
2. Assign each edge (i, j) the weight $I(x_i, x_j)$. BIC
3. Start from an empty graph and iteratively add the edge with the highest weight.
4. Skip any edge that creates a cycle.
5. Continue until all nodes are connected.

This yields the globally optimal tree under the maximum likelihood criterion.



Question 4: [15 marks]

Consider the following directed PGM



where each random variable is Boolean-valued (True or False).

CPT

(a) Write the format (with empty values) of the *conditional probability tables* for this graph. **[5 marks]**

(b) Suppose we observe n sets of values of A, B, C (complete observations). The *maximum-likelihood principle* is a popular approach to training a model such as above. What does it say to do? **[5 marks]**

(c) Suppose we observe 5 training examples: for (A, B, C) — (F, F, F); (F, F, T); (F, T, F); (T, F, T); $(T, T; T)$. Determine *maximum-likelihood estimates* for your tables. **[5 marks]**

a).

P(A=True)
—

P(B=True)
—

A	B	Pr(C=T|A,B)
T	T	—
T	F	—
F	T	—
F	F	—

$x_1 \cdots x_n$

b). $\underset{\text{tables parameters}}{\text{Argmax}} \prod_{i=1}^{n} P(A=a_i)\,P(B=b_i)\cdot \Pr(C=c_i \mid A=a_i, B=b_i)$

c). $P(A=\text{True}) = \frac{2}{5}$ $\qquad P(B=\text{True}) = \frac{2}{5}$

$\Pr(C=T \mid A=T, B=T) = \frac{1}{1} = 1$

$\Pr(C=T \mid A=T, B=F) = \frac{1}{1} = 1$

$\Pr(C=T \mid A=F, B=T) = \frac{0}{1} = 0$

$\Pr(C=T \mid A=F, B=F) = \frac{1}{2} = \frac{1}{2}$

Extra: Not Compulsory, Go through to understand better

Exercise: Learn a Maximum Spanning Tree via Mutual Information (Chow–Liu)

MRF Tree

Data. We observe $T = 5$ i.i.d. samples of $n = 4$ binary variables $A, B, C, D \in \{0, 1\}$:

Sample	A	B	C	D
1	0	0	0	0
2	0	0	1	0
3	1	1	1	1
4	1	1	0	1
5	1	1	1	1

Goal. Assume the distribution over (A, B, C, D) is tree-structured. Learn the *maximum spanning tree* using pairwise **mutual information** (MI) as edge weights.

Mutual Information (definition). For any pair (X_i, X_j) with joint pmf $P_{ij}(x_i, x_j)$ and marginals $P_i(x_i), P_j(x_j)$,

$$I(X_i; X_j) = \sum_{x_i} \sum_{x_j} P_{ij}(x_i, x_j) \log \frac{P_{ij}(x_i, x_j)}{P_i(x_i) P_j(x_j)}.$$

(Any log base is fine; use natural log unless stated.)

Empirical estimation. From the table, compute counts $\hat{n}_{ij}(x_i, x_j)$ and $\hat{n}_i(x_i)$, and estimate

$$\widehat{P}_{ij}(x_i, x_j) = \frac{\hat{n}_{ij}(x_i, x_j)}{T}, \qquad \widehat{P}_i(x_i) = \frac{\hat{n}_i(x_i)}{T}.$$

Then the empirical MI is

$$\widehat{I}(X_i; X_j) = \sum_{x_i} \sum_{x_j} \widehat{P}_{ij}(x_i, x_j) \log \frac{\widehat{P}_{ij}(x_i, x_j)}{\widehat{P}_i(x_i) \widehat{P}_j(x_j)}.$$

Tasks.

1. Compute $\widehat{I}$ for all $\binom{4}{2} = 6$ pairs: (A, B), (A, C), (A, D), (B, C), (B, D), (C, D).
2. Use $\widehat{I}$ as edge weights and run Kruskal/Prim to obtain the maximum spanning tree.
3. Report the selected edges.

Worked Example: Compute One Mutual Information Step

Consider variables A and B from the data:

A , B

算互信息·process

Sample	A	B
1	0	0
2	0	0
3	1	1
4	1	1
5	1	1

1. Joint and marginal empirical distributions.

(A,B)	$(0,0)$	$(0,1)$	$(1,0)$	$(1,1)$
count	2	0	0	3
$\hat{P}_{AB}(a,b)$	0.4	0	0	0.6

Marginals:

$$\hat{P}_A(0) = 0.4, \quad \hat{P}_A(1) = 0.6, \quad \hat{P}_B(0) = 0.4, \quad \hat{P}_B(1) = 0.6.$$

2. Mutual information formula.

0.6

0.6

$$\boxed{\hat{I}(A;B)} = \sum_{a,b} \hat{P}_{AB}(a,b) \log \frac{\hat{P}_{AB}(a,b)}{\hat{P}_A(a)\hat{P}_B(b)}.$$

0.6 0.6

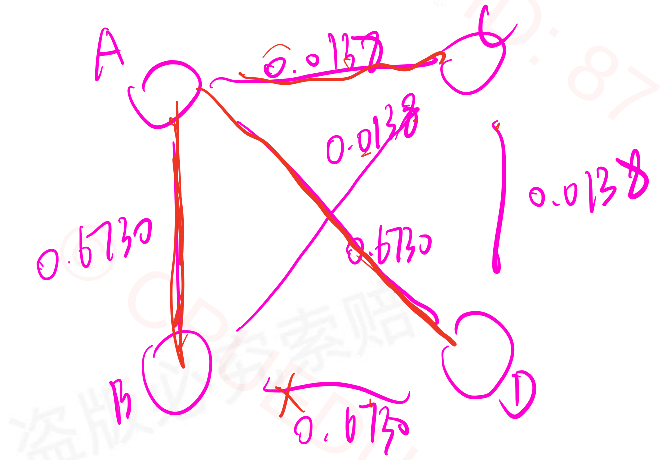
Only nonzero pairs $(0,0)$ and $(1,1)$ contribute:

$$\hat{I}(A;B) = 0.4 \log \frac{0.4}{(0.4)(0.4)} + 0.6 \log \frac{0.6}{(0.6)(0.6)}$$
$$= 0.4 \log(2.5) + 0.6 \log(1.667) = 0.673 \text{ nats}.$$

$C_4^2 = 6$对

3. Empirical mutual informations (summary).

pair	(A,B)	(A,C)	(A,D)	(B,C)	(B,D)	(C,D)
$\hat{I}$	0.6730	0.0138	0.6730	0.0138	0.6730	0.0138



Question 6: [20 marks] PGM Modeling

Your task is to design a system for alerting residents of the Dandenongs that they should evacuate for an impending bushfire. The Dandenongs is an area of Victoria that suffers from regular bushfires in the summer when temperatures are high, and humidity low. When fires are close to a fictional town called Bayesville, you must alert residents that they should evacuate. If fires are too close, then you should not advise evacuation as residents are safer if they stay where they are (at home).

The Country Fire Association has deployed sensors around the area that monitor whether a fire is in progress at each sensor's location; in particular if any of three sensors S_1, S_2, S_3 are 'on' then residents should evacuate. However if either of the closer sensors C_1, C_2 are 'on' then residents should stay put.

(a) Model the above problem as a *directed probabilistic graphical model* (PGM). In particular, you need not provide any probabilty tables, just the graph relating random variables S_1, S_2, S_3, C_1, C_2 and an additional r.v. A for alerting residents to evacuate. **[5 marks]**

(b) How many *conditional probability tables (CPTs)* should be specified for your model, and what should these tables' dimensions be? **[4 marks]**

(c) Suppose you are told by the Fire Commissioner that the sensors are always accura[illegible] you say about your CPTs? **[4 marks]**

(d) How would you change your model if the Commissioner then tells you that the sensor S_1 is not perfectly accurate? **[4 marks]**

(e) Given this final model, assuming you have trained it and completed all the necessarily CPTs, how would you use it to drive the alarm to evacuate? **[3 marks]**

a).

Add r.v.
F_1, O_1

d).

b). 6 , 其中5个table只有一个列

$Pr(S_i=T)$

第6个table

6列

$2^5=32$行

S_1	S_2	S_3	C_1	C_2	$Pr(A=T \mid S_1,S_2,S_3,C_1,C_2)$
1	0	0	0	0	1
1	0	0	1	0	0

e. $P(A=True)$

use elimination variable to get $P(A=T)$

c). 当所有sensor准确，代表第6个CPT是确定性的

$(S_1 \vee S_2 \vee S_3) \wedge \neg(C_1 \vee C_2)$

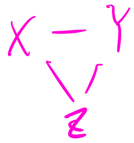
但对前5个CPT没作用

(d) Can the independence assumptions of all *directed probabilistic graphical models* be represented by *undirected probabilistic graphical models*? Why or why not? [5 marks]

No. 如

$X \perp Y$, $X \not\perp Y \mid Z$

Equivalence: 独立性是否相同



相互依赖

(d) Can the joint probability distribution described by any *undirected probabilistic graphical model* be expressed as a *directed probabilistic graphical model*? Explain why or why not. [5 marks]

3分

Yes, 因为所有联合分布都能被 chain Rule factorization 来表达

Eg. $P(A,B,C,D) = P(A) \cdot P(B \mid A) \cdot P(C \mid A,B) \cdot P(D \mid A,B,C)$

CPU EDUCATION